

Sample Paper 6 — Class 5 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 50 questions / 60 marks / 60 minutes / no negative marking as Papers 1–5, but noticeably harder throughout — closer to SOF IMO **Level 2**. Attempt this only after scoring comfortably on Sample Paper 5.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Whale B) Dolphin C) Bat D) Shark
- Q2.** Author is to Book as Composer is to _____. A) Music B) Song C) Instrument D) Concert
- Q3.** If each letter of a word is moved 2 steps forward in the alphabet, MUSIC becomes OWUKE. Using the same rule, how is PAINT written? A) RCKPV B) RCKPU C) RCKOV D) SCKPV
- Q4.** This series follows two interleaved patterns — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 2, 100, 4, 97, 8, 94, 16, ____ A) 88 B) 89 C) 90 D) 91
- Q5.** Five friends sit in a row. Bala sits exactly in the middle. Aman sits at the left end. Chetan sits between Aman and Bala. Esha sits between Bala and Dinesh. In what order are they seated, left to right? A) Aman, Chetan, Bala, Esha, Dinesh B) Aman, Bala, Chetan, Dinesh, Esha C) Chetan, Aman, Bala, Dinesh, Esha D) Aman, Chetan, Bala, Dinesh, Esha
- Q6.** Priya walks 9 m North, then 7 m East, then 9 m South, then 7 m West. Where is she now relative to her starting point? A) 9 m North B) 7 m East C) Back at the starting point D) 16 m away
- Q7.** If the 5th day of a month is a Tuesday, what day of the week is the 26th of the same month? A) Monday B) Tuesday C) Wednesday D) Saturday
- Q8.** Find the odd one out: A) 27 B) 64 C) 125 D) 150 (Hint: three of these follow the same multiplication pattern — a number multiplied by itself twice.)
- Q9.** Find the next number: 2, 6, 24, 120, ____ A) 360 B) 480 C) 600 D) 720
- Q10.** In a row, Neha is 7th from the left and 12th from the right. If two more children join the row at the left end, what will be her new position from the left? A) 7th B) 8th C) 9th D) 10th
- Q11.** Find the odd one out: A) 17 B) 34 C) 51 D) 67
- Q12.** In a certain code, EARTH is written as GCTVJ (each letter moved 2 steps forward). How is PLANET written in the same code? A) RNCPGV B) RNCPGU C) RMCPGV D) SNCPGV
- Q13.** Find the odd one out: A) Square B) Rhombus C) Trapezium D) Cube

- Q14.** In a competition, the second-highest-scoring team scored more than Team X but less than Team Y. Team Z scored the highest overall. Given that Team Y is not the highest, who scored second-highest? A) Team X B) Team Y C) Team Z D) Cannot be determined
- Q15.** Study the pattern: 1, 4, 10, 20, 35, _____. Find the next number. A) 50 B) 52 C) 54 D) 56

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $4,56,789 + 2,87,654 = ?$ A) 7,44,443 B) 7,44,343 C) 7,43,443 D) 7,45,443
- Q17.** $9,00,000 - 4,56,789 = ?$ A) 4,43,211 B) 4,43,111 C) 4,42,211 D) 4,44,211
- Q18.** $6,789 \times 14 = ?$ A) 95,046 B) 94,046 C) 95,946 D) 95,146
- Q19.** $98,784 \div 24 = ?$ A) 4,116 B) 4,106 C) 4,126 D) 4,216
- Q20.** Find the HCF of 72 and 96. A) 12 B) 16 C) 24 D) 48
- Q21.** Find the LCM of 15 and 20. A) 40 B) 60 C) 80 D) 100
- Q22.** Which of these fractions is the greatest: $\frac{5}{7}$, $\frac{3}{4}$, $\frac{7}{9}$, $\frac{8}{11}$? A) $\frac{5}{7}$ B) $\frac{3}{4}$ C) $\frac{7}{9}$ D) $\frac{8}{11}$
- Q23.** $2 \text{ km } 450 \text{ m} + 1 \text{ km } 800 \text{ m} = ?$ A) 3 km 250 m B) 4 km 250 m C) 4 km 50 m D) 3 km 950 m
- Q24.** A rectangular field is 24 m long and 18 m wide. Find its area. A) 408 sq m B) 420 sq m C) 432 sq m D) 442 sq m
- Q25.** A regular pentagon has a perimeter of 60 cm. Find the length of each side. A) 10 cm B) 11 cm C) 12 cm D) 13 cm
- Q26.** The cost price of an item is ₹1,200 and the selling price is ₹1,560. Find the profit percentage. A) 25% B) 28% C) 30% D) 32%
- Q27.** A sum of money doubles itself in 8 years at simple interest. Find the rate of interest per annum. A) 10% B) 12.5% C) 15% D) 20%
- Q28.** ₹360 is divided among three people in the ratio 5:6:7. Find the largest share. A) ₹120 B) ₹130 C) ₹140 D) ₹150
- Q29.** Convert 0.125 to a fraction in simplest form. A) $\frac{1}{8}$ B) $\frac{1}{4}$ C) $\frac{1}{5}$ D) $\frac{1}{10}$
- Q30.** $4 \text{ kg } 500 \text{ g} + 2 \text{ kg } 750 \text{ g} + 1 \text{ kg } 800 \text{ g} = ?$ A) 8 kg 50 g B) 9 kg 50 g C) 9 kg 950 g D) 8 kg 950 g
- Q31.** How many seconds are there in 2.5 hours? A) 8,000 B) 8,500 C) 9,000 D) 9,500
- Q32.** Find the perimeter of an equilateral triangle with side 23 cm. A) 46 cm B) 69 cm C) 92 cm D) 23 cm
- Q33.** Find the area of a rectangle with length 35 cm and breadth 24 cm. A) 820 sq cm B) 830 sq cm C) 840 sq cm D) 850 sq cm
- Q34.** How many edges does a triangular prism have? A) 6 B) 8 C) 9 D) 12

Q35. Find the number of diagonals in a hexagon. A) 6 B) 8 C) 9 D) 12

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

Q36. A school collected ₹1,84,000 from 8 classes equally. How much did each class contribute? A) ₹22,000 B) ₹22,500 C) ₹23,000 D) ₹23,500

Q37. A factory makes 2,160 bottles in 9 hours, at a steady rate. How many bottles does it make in 5 hours? A) 1,150 B) 1,200 C) 1,250 D) 1,300

Q38. A tank holds 3,600 L and is filled by a pipe at 200 L per minute. How long will it take to fill the tank completely? A) 16 min B) 17 min C) 18 min D) 20 min

Q39. A bus leaves city A at 7:15 a.m., stops for 20 minutes on the way, and reaches city B at 11:50 a.m. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 5 min B) 4 h 15 min C) 4 h 25 min D) 4 h 35 min

Q40. A chemist mixes 3 L 250 mL of one liquid with 1 L 750 mL of another, then pours the mixture equally into 4 bottles. How much does each bottle get? A) 1 L B) 1 L 250 mL C) 1 L 500 mL D) 1 L 750 mL

Q41. Rohit buys 8 notebooks at ₹32 each and gets a discount of ₹26 on the total bill. How much does he pay? A) ₹220 B) ₹225 C) ₹230 D) ₹235

Q42. A cistern with 1,200 L of water is used for irrigation at the rate of $\frac{1}{8}$ of its original capacity every day. How much water remains after 3 days of use? A) 700 L B) 725 L C) 750 L D) 775 L

Q43. A delivery van carries 4 boxes weighing 22 kg 500 g, 18 kg 750 g, 15 kg 250 g, and 9 kg 500 g. Find the total weight carried. A) 65 kg B) 66 kg C) 67 kg D) 68 kg

Q44. 1,520 chairs are arranged equally into rows of 38 chairs each. How many rows will there be? A) 36 B) 38 C) 40 D) 42

Q45. A trader bought 80 kg of rice for ₹3,200 and sold all of it at ₹48 per kg. Find his total profit. A) ₹560 B) ₹600 C) ₹640 D) ₹680

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

Q46. A number is first multiplied by 6, then 24 is subtracted from the result, giving 108. Find the original number. A) 20 B) 21 C) 22 D) 23

Q47. The sum of three numbers is 150. The second number is twice the first, and the third is 10 more than the second. Find the first number. A) 24 B) 26 C) 28 D) 30

Q48. Anjali had a certain amount of money. She spent $\frac{2}{7}$ of it on books, then $\frac{1}{5}$ of what remained on stationery, and was left with ₹1,000. How much money did she have at first? A) ₹1,500 B) ₹1,750

C) ₹2,000 D) ₹2,250

Q49. In a school, the number of girls is 8 more than twice the number of boys. There are 308 students in total, and 100 of them are boys. Find the number of girls. A) 198 B) 200 C) 204 D) 208

Q50. A sequence starts with 3 and 5, and each later term is the sum of the two terms before it: 3, 5, 8, 13, 21, _____. Find the next term. A) 29 B) 31 C) 34 D) 36

Answer key: [sample-paper-06-answer-key.md](#)
